

## Assignment - OS.

Q1.

→ A user-defined function is a function is a `func()`, created by user programmer using `def` keyword.  
Used to make code re-usable.

Q2.

→ A) function w parameters :

- Accepts value as input
- More flexible & reusable.

eg- `def Add (iNo1, iNo2) :`  
    `return (iNo1 + iNo2)`  
`Add (10, 11)`

B) function w/o parameters :

- Does not accept any input
- Performs a fixed task

eg- `def display () :`  
    `print ("Hello!")`  
`display()`

Q3.

→ Output: 10  
NameError : 'x' is not defined.

Q4.

→ `def message () :`  
    `print ("welcome")`

`message()`

Q5

→

A] print()

→ Displays output on the screen, not re-usable

B] return()

→ Sends a value back to the caller, this (returned) can be saved in & re-used.

Q7.

→

Output: <class 'int'>

<class 'float'>

<class 'str'>

} stored in a.  
→ shows dynamic typing

Q8. Q6

→

$x = 100$ , works without declaration, as python + automatically creates the variable when a value is assigned. There is no need to declare its data type.

Q9.

→

$x = 10$

$x = \text{"TEN"}$

} → here first  $x$  holds an int but later it stores a string.

→ Possible due to dynamically typed lang.

Q10.

→

Python automatically allocates memory when object are created & frees memory when they are no longer needed using Garbage Collection.

Thus we don't need to manually allocate / free memory reducing memory leaks → Easy.